RevTrace: Preserving Valid Progress across Trusted Feedback Revisions

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***Abstract*—Trusted feedback can withdraw a timely transcript even when its computation remains valid. RevTrace preserves such progress by separately confirming complete call inputs, retaining original selection authority, and checking support for the Consumer's whole view. A conditional call can remain eligible even when its own next prediction is wrong; an accepted result can outlive its producer. A selected-trace refinement argument connects this coordination to the protected program. In a predeclared 24-speaker stratum, computation retention alone increases frontier withdrawals from 12 to 17; joint retention yields zero and adds 26.645 seconds of deadline credit over 714.110 seconds offered. In a matched native correction study, the final protocol preserves original histories, delivers 65.822 of 66.340 deadline-credited seconds, and passes both whole-run deadlines in six of eight trials, versus none for conservative execution. Fault and sustained-state tests expose recovery, quality, and resource boundaries.**

***Index Terms*—dependable inference, feedback state, incremental computation, speculative execution, streaming ASR**

I. Introduction

A streaming transcript can arrive on time and still disappear when trusted feedback advances. In one actual run, a complete two-second audio interval introduced 37 visible characters at 1.444 s, within its two-second display deadline. A trusted update at 1.544 s replaced that whole hypothesis with a shorter snapshot, although its computation remained compatible with the selected feedback state. Identical text returned through certified delivery at 3.720 s, within the eight-second deadline. Both first-delivery obligations were met, yet the phrase was absent for 2.176 s, including at the original display deadline (Figure 1). The service lost delivered progress without needing to correct it. Trusted feedback here means feedback accepted by protected selection.

This problem arises when generated or corrected output becomes an input to subsequent inference. Streaming speech correction is a concrete case: StreamCorrect feeds its chosen correction into later recognition [1]. A new trusted selection can invalidate some speculation while confirming other work that began earlier. Recomputing everything is conservative, but retaining computation alone is also insufficient. In our predeclared 24-speaker evaluation, computation retention improves timely delivery while increasing audio-frontier withdrawals from 12 to 17. Jointly retaining computation and its supported whole view yields zero. The missing obligation is to turn valid work into *sustained service* across a trusted revision.

Three questions determine the actions. Does an old call's complete frozen input still match the selected parent feedback state? Has the original selection authorized its result? Does the Consumer's current whole hypothesis still have a complete dependency path from that parent? These questions concern different objects. A call can have a confirmed input while its own next prediction is wrong. A view can lose support while a future call's input is unresolved. A producer can die after its complete result has already been accepted. A birth version, matching text, or producer-liveness bit cannot stand in for all three judgments.

RevTrace checks complete frozen inputs and dependency paths before retaining work, leaves result selection to the original program, and revalidates the Consumer's entire displayed hypothesis when a selected state installs. Incremental dialogue and computation supply established dependency and revocation principles [2]–[4]. SSBD already combines draft reuse and display masking [5]; PredGen speculates across evolving speech stages [6]. Our focus is a recursively selected feedback task: retained work must still execute its original correction and update, and applying a new selected root must separately validate the actual whole view. The contribution is this concrete coordination contract and its service consequences, rather than reuse or revocation in isolation.

We make three contributions:

- We identify a service gap between timely first delivery and continued availability under compatible feedback, and define executable witnesses for complete-call eligibility and whole-view retention (Sections II–III).
- We implement their joint coordination with original selection, durable facts, and physical ownership. A conditional selected-trace refinement argument explains how early full calls preserve the protected task, including accepted results across supported process failures (Sections III–IV).
- We test the composition on real neural service: factorial controls isolate the interaction, native correction tests original full calls, and fault and sustained-state controls establish its costs and limits (Section V).

Deadline credit counts offered audio seconds meeting both first-display and certified-delivery deadlines. The headline joint arm adds 26.645 s over 714.110 s offered with unchanged selected histories. A matched native comparison raises credit from 30.0 to 65.822 of 66.340 s and whole-run passes from zero to six of eight; resident independent requests pass seven of eight. The remaining EOS misses, overload, cold recovery, and no-reset quality failures delimit the service claim.

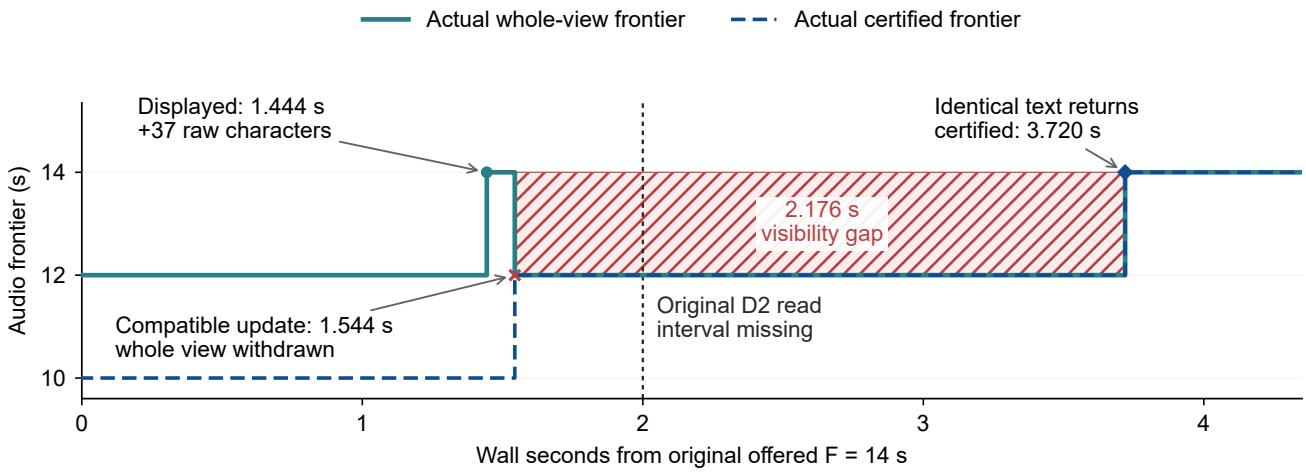


Fig. 1. Timely first delivery can still leave a gap in delivered progress. An actual computation-only run displays a complete interval at 1.444 s, withdraws it under a compatible update at 1.544 s, and restores identical text at 3.720 s. Shading marks the 2.176-s gap; the vertical line is the original D2 read point. The view introduces 37 raw code points including spaces, not character/audio alignments. This c16 overload case illustrates the failure mode, not its incidence or a whole-run service win.

II. Feedback Contract

A. Feedback creates an execution dependency

An immutable audio stream supplies a protected selection path producing *Canonical* facts and a faster *Private* path computing and displaying provisional hypotheses. Selection can change the next call's input: StreamCorrect feeds its chosen correction into subsequent ASR [1]. We coordinate this dependency as the paths advance asynchronously.

For a fixed session and execution contract, write an installed fact as

$$C = (v, F, Z_C, P_C, T_C). \tag{1}$$

Here v is the version, F the certified processing frontier, Z_C the complete selected output, P_C the feedback state, and T_C its bound text. The contract fixes the model, adapter, decoding, audio identity, and transformations to P and T. Consumer coverage follows the adapter's publication policy; advancing F need not add characters. Certification means acceptance by protected selection and installation, separately from human-transcription accuracy.

The adapter declares a conservative representation of *all mutable cross-call state that can affect later model inputs, selection, or feedback*. Telemetry and resources are separate. Qwen passes prompt tokens without cross-call numerical KV caches. The native adapter additionally carries audio buffers, token groups, context, alignment position, and decoder configuration. Comparisons require the same contract: text equality cannot replace an omitted cursor or numerical cache. The representation may conservatively include fields unnecessary to a particular call.

B. Three lifetimes of provisional work

A complete provisional node n records a frozen call

$$q_n = (a_n, P_n^-, b_n, \chi_n), \quad n = (q_n, Z_n, P_n^+, T_n), \tag{2}$$

where a_n is the input-state frontier, $b_n > a_n$ the audio target, and χ_n binds remaining inputs and execution identity. The model may read the prefix through b_n; a_n does not assert token-to-audio alignment. The adapter derives P_n^+ and T_n from a complete result. An in-flight call has only its frozen input; partial or cancelled tokens supply no node.

The *dependency lifetime* determines usability under the current root; the *visible lifetime* determines whole-hypothesis retention; the *physical lifetime* ends only when execution finishes or drains. A new root can change the first two before the third ends. Their decisions therefore require separate evidence.

The service retains every accepted input obligation until certified processing covers it. Discarding a branch, requesting cancellation, or recovering an old view does not remove that obligation or reset its arrival clock. In exact mode, a missed deadline never authorizes an uncertified Canonical transition.

C. Trust and failure boundary

We assume one trusted Owner per session, protected selection, immutable audio and model configuration, unambiguous encoding, collision-resistant digests, and storage honoring successful synchronization. Used calls must faithfully consume their frozen inputs. Host receipts bind API objects, not GPU arithmetic; independent Private and Authority floating-point executions need not agree.

The interface checks identity, completeness, derived state, and ancestry. Native resilience supports worker or Owner-group fail-stop on an intact filesystem; Supervisor, accepted-input ledger, actual Consumer, power observer, and synchronized storage survive. Their original input clocks continue through restart. Supervisor/Consumer failure, host power loss, compromised Owner or Authority, Byzantine code, persistent corruption, and lost storage are excluded. Controlled wrong-input injections in Section V intentionally violate normal Private input fidelity;

we use them to study containment and executor release, not as examples covered by the lawful-continuation proposition.

III. Valid Continuations and Whole Views

A. A continuation under a new root

The central question is whether an old call still belongs to a valid provisional continuation after trusted feedback advances. Its birth version records dispatch history. The dependency witness establishes whether it remains usable now.

Let $x = (F, P_C)$. A complete recorded node provides an edge

$$(F, P_C) \xrightarrow{n} (b_n, P_n^+) \quad \text{if} \quad a_n = F \ \wedge \ P_n^- = P_C, \tag{3}$$

under the input, identity, and execution contract of Section II. A finite sequence of such edges is a *valid provisional continuation* of C when each successor consumes the preceding complete output state at its actual frontier. Write $C \rightsquigarrow n$ for a complete path to n. A frozen in-flight successor can belong to the same continuation, but contributes no usable result until completion.

Proposition 1 (Sufficient re-rooting witness)**.** *Suppose an old branch contains a complete boundary node at the new certified frontier F, with output state P_C. If its subsequent recorded calls form an unbroken input–output path to the actual branch tip, those nodes are a valid provisional continuation of C. An in-flight successor may continue when its unchanged frozen input is that tip and its target exceeds F.*

Proof. The boundary equals (F, P_C), so the first successor satisfies Eq. (3). Each subsequent call consumes the preceding complete output at its recorded frontier under the same contract. Induction reaches the tip and the unchanged input of its in-flight successor, which supplies no output yet. □

This establishes dependency validity, not Canonical write authority. Provisional results remain revisable. The witness follows actual calls without assuming bitwise equality of different floating-point executions.

A branch born under v can reach (F, P_C) when $v+1$ installs, making version-only rejection wasteful. Conversely, equal text can hide different tokens or alignment state; equal state at a different frontier also fails Eq. (3).

B. Retaining computation: R

The core Qwen guard conservatively implements Proposition 1. On installation, it requires a complete Private node at exactly F with full feedback tokens P_C, then checks frozen inputs and frontiers along complete successors to the actual tip. An intermediate match is insufficient. An in-flight successor must be the unique tracked task, consume that tip, and target audio beyond F.

Success retains the branch without changing job IDs, birth versions, inputs, or results; otherwise the Owner fences it and rebuilds from an allowed state. Boundary-node and unique-slot requirements restrict this sufficient guard, not every valid reuse scheme.

C. Retaining an actual view: S

A snapshot Consumer can replace a valid ahead view with a shorter certified snapshot despite computation retention. RevTrace maintains separate certified and provisional whole-hypothesis lanes, without an uncertified suffix splice or certified-prefix status for retained views.

For a Canonical application, S checks the complete visible node's input directly against (F, P_C) or follows complete published ancestors to that boundary. Missing, cyclic, or ambiguous ancestry is rejected. Unlike R's conservative guard, a directly rooted published call needs no extra Private boundary node.

Before actual application and its receipt, the kernel checks the preview against the durable root, registered results, dependency path, and bound text/frontier. A caller-supplied compatibility flag cannot replace these records.

Proposition 2 (Continuity at a compatible application)**.** *Immediately before an authorized Canonical application, let the visible whole provisional view h be ahead of its new frontier F. If S proves $C \rightsquigarrow h$ and no independent withdrawal mode is invoked, that application advances the certified lane and preserves h unchanged.*

Proof. The authorized compatible-and-ahead branch installs C in the certified lane while retaining h's node, text, and frontier. That application therefore withdraws none of h. □

Incompatible updates must withdraw the view, and later valid results can revise it. This is a per-application property, not global monotonicity. Snapshot consumption and conservative recomputation can be safe; the four-arm study measures the extra progress and cost of joint retention.

D. Native feedback and conditional full calls

StreamCorrect supplies a concrete recursive feedback task [1]. For complete closed state P and accepted audio X, expose its recognition boundary as

$$A(P, X) = (B, M), \tag{4}$$

where B contains ordered beam candidates and M the engine state and inputs needed by original correction and update (C/U). Candidate order, prompts, returned token IDs, selection, and update rules remain intact.

Before learned correction finishes, the original update tail with a typed first-beam identity prediction produces provisional closed state $\widehat{P}^+$. It is a dependency projection, not a complete selected result. A second worker can run the next *complete original call* on that conditional input; both GPUs host the full recognizer–corrector. Its own prediction may be wrong even when its input later matches. Original C/U, rather than the prediction, determines its output.

When the parent installs, the Owner compares the call's complete frozen start state, actual frontier, contract, and session epoch with it. Equality confirms the unchanged owned call: it continues if running, or becomes eligible for ordered selection

TABLE I

COORDINATOR FOR NATIVE CONDITIONAL FULL CALLS. INPUT STATUS AND WHOLE-VIEW SUPPORT ARE SEPARATE JUDGMENTS; SELECTION STILL USES THE ORIGINAL COMPLETE RESULT. THIS SUMMARIZES EXISTING GUARDS, NOT A NEW SCHEDULER.

State: selected cut C; owned calls J; accepted complete buffers B; selected same-frontier witnesses W; actual whole view h.
Input status $E(j)$: exact current parent and all frozen bindings match: CONFIRMED; actual W at j's start differs: REFUTED; no such witness: UNKNOWN. Covered targets and fenced sessions are ineligible.
1. **Dispatch:** prefer missing selected work on a ready worker; otherwise freeze an unduplicated conditional full call within the ahead bound. Keep original UID, birth, incarnation and accepted-input clocks.
2. **Receive:** check channel, identity, frozen input and complete original output. Local duplicate/fenced rejection cannot clear another lease. Reject refuted, covered or fenced results; buffer an admissible complete result if its input remains unknown.
Typed ASR projections extend only a checked provisional path; they do not select.
3. **Select ready:** for a complete original result with $E(j) =$ CONFIRMED, recheck the installed parent, synchronize its bound fact, then install C and record W. Preserve original identity; reclassify pending results. Unknown grants no commit authority.
4. **Consume:** check fact order and actual parent. Advance the certified lane; retain an ahead h whole when $S(C, h)$ proves its path. Otherwise render the installed cut, separately from deciding unresolved calls.
5. **Release:** actual consumption releases outbox references, not live-view references. Physical ownership ends at the matching terminal or confirmed exit/cleanup; cancellation acknowledgment is insufficient.

if complete. Job ID, birth, producer incarnation, inputs, and execution history stay unchanged.

An earlier ancestor mismatch can remove current view support without refuting a later complete input. Cancellation before correction requires a disagreeing actual selected state at that call's own start frontier, coverage of its target without remaining use, or a changed epoch. An unresolved future parent supplies none of these proofs. The view can therefore require withdrawal while the call remains unresolved.

This native input witness and the core Qwen boundary-node/unique-successor guard are different sufficient implementations of the continuation relation. Neither grants write authority. Both use a whole-view Consumer that keeps certified and provisional provenance separate. The native comparison holds two complete workers fixed and varies conditional dispatch and retention; ready-only ASR handoff is a separate developmental control.

Table I summarizes the native event loop. An unknown future input authorizes neither selection nor a refutation-based cancel. When a complete result replaces its typed prediction, only the suffix that consumes the actual output contiguously survives. Worker loss is handled separately: validate complete sealed-channel receipts before fencing new old-incarnation messages; do not erase already accepted B. Thus losing a view, rejecting a result, and releasing a worker are different actions, even when triggered by one feedback event.

E. Preserving the protected program's semantics

Let $P \Rightarrow_{\mathcal{A},X} P'$ denote a transition allowed by the original protected program on accepted immutable input X. The relation may be nondeterministic. A complete original call includes its original selection and update, or its declared buffer-only path; a typed identity projection is not such a result. We project a protocol execution onto its distinct durable selected facts, deleting speculation, waiting, local rejection, cancellation, collection, and repeated delivery. This restricted projection follows the standard stuttering view of refinement [7]; provisional whole-view behavior remains outside its alphabet and has a separate compatibility property.

Proposition 3 (Selected-trace refinement)**.** *Assume sufficient declared state, faithful execution of frozen inputs, exclusive serial selection, correctly bound message identities, and storage that honors successful synchronization. If each new durable fact binds a complete original call whose full input equals the preceding selected cut, retains the call's original identity, was admitted under valid message authority, and is synchronized before installation or certified application, then the projected facts form an execution allowed by $\mathcal{A}$. Replaying an existing fact adds no transition.*

Proof. The initial cut belongs to $\mathcal{A}$. Deleted actions do not extend the prefix. Each new fact binds a complete original call on exactly the preceding cut and accepted input, hence an allowed next edge regardless of dispatch time. Synchronization precedes privileged application. Recovery replays the same checked identities, including a fact first consumed after an Owner crash, without adding an edge. Induction proves the claim. □

The durable append is the selection boundary; installation and consumption do not select again. Already accepted buffers retain their producer identities after exit, whereas new fenced messages are inadmissible. The certified frontier derives only from checked facts; ahead views remain provisional. Input confirmation does not certify predicted next states.

A separately frozen specification explores 6,231 two-frontier normal states and 14,074 one-frontier failure states, each to a 12-event depth, with 27 named controls. An independent trace checker binds actual facts and whole-view applications to original call receipts. The specification is not extracted from the runtime; depth bounds truncate exploration, and neither check verifies all Python, CUDA, or operating-system behavior. Exact equality with a fixed neural reference is separate empirical evidence, rather than a premise that arbitrary GPU reruns must be bitwise identical.

IV. EXECUTION PROTOCOL

A. Complete results and physical ownership

In the core runtime, dispatch binds session, incarnation, job, input, target, and contract. A row returns either a complete result with re-derived state/text or cancellation without partial state. Complete rows can be installed and consumed before

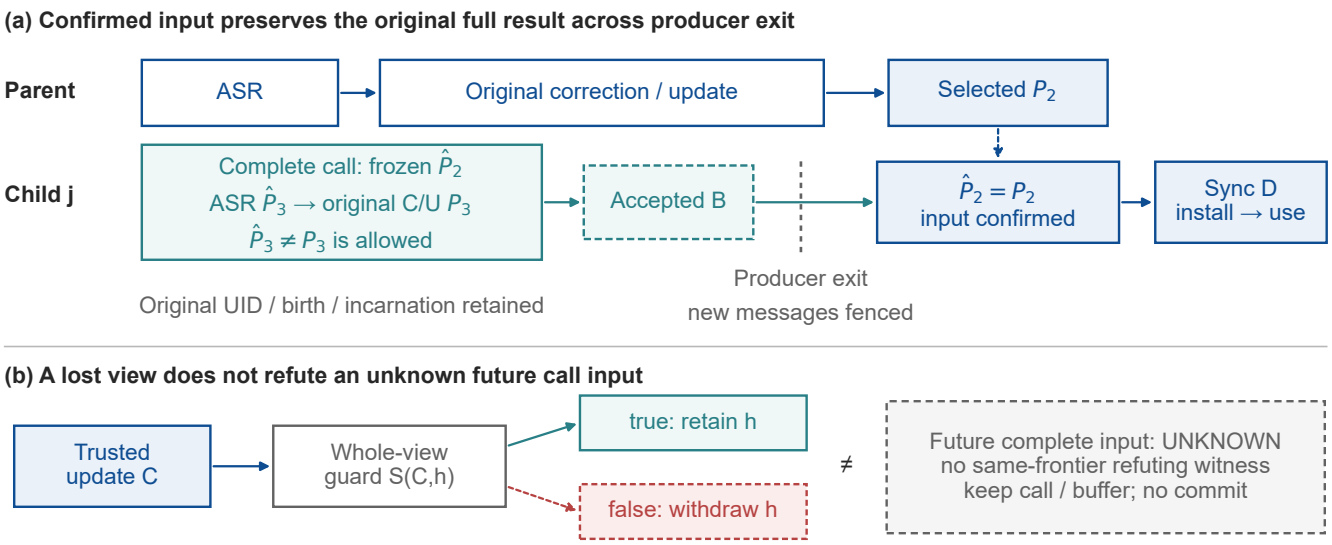


Fig. 2. Eligibility, original selection, and whole-view support are separate decisions. (a) A complete child call runs on frozen $\widehat{P}_2$; its own prediction $\widehat{P}_3$ may differ from original C/U output P_3. An accepted buffer keeps its original identity across producer exit and fencing. Matching selected P_2 confirms the input; durable sync precedes installation and use. (b) Losing view support does not refute an unknown future input or authorize its commit. This is a causal native schematic, with complete-state and execution assumptions from Section II, not a measured timeline.

their batch finishes. Only all terminal rows and device synchronization authorize `BatchDrained` and executor reuse. The native process lifecycle below has separate evidence.

B. Covered work and safe-point exit: K

For the tested use model, K requests exit when the installed Canonical frontier covers a proposal's target and it has no remaining declared use. This differs from R: valid work beyond the new root can remain useful. Workers observe `CancelRequest` at decode boundaries; completion can win, but each row has one terminal outcome. Static batches are not compacted or refilled; masked costs and drain delays remain measured.

C. Durable facts and outstanding input

Before acknowledgment, synchronized frames bind parent/next states, request identity, selected complete result, and receipt; recorded deltas must reconstruct the state. Group-ready commit shares synchronization among validated sessions without a collection timeout or cross-session atomic transaction. Uncertain append/application outcomes block writes. Write-ahead recovery [8] reconstructs facts and identities; the native extension binds this to original correction calls.

D. Validation boundaries

Propositions 1 and 2 concern the stated guards and compatible applications. The artifact retains an earlier, separately frozen recovery prototype and its bounded model and crash results; they do not validate the later core or native fault paths. The selected-trace argument of Proposition 3, its finite specification, and actual execution checks have distinct scopes.

E. Continuing through delivery and process failures

Each job has an Owner namespace and monotonic issue number; each physical slot has a worker incarnation, using established RPC identity principles [9]. Fencing removes future message authority without renaming already accepted complete results.

a) Local rejection and physical release: The ingress guard rejects exact duplicates and fenced messages locally; they cannot mutate selected state or release newer leases. Contradictory currently owned contents freeze writes. Per-incarnation Pipes and receiver threads isolate incomplete frames. Replacement requires process exit, CUDA release and joined receivers. Strictly validate complete receipts from the sealed channel before fencing the incarnation and retiring unfinished work. Accepted buffers and supported views retain their original identities.

b) Durable selection and recovery: A synchronous hook binds complete parent/next states, original correction and IDs, job, Owner/session epochs, and producer incarnation. Framing, checksums and chaining protect the fact; synchronization precedes installation/publication. Uncertain writes freeze the writer. Recovery discards only an incomplete final frame; complete corruption fails. These conventional write-ahead rules [8] bind the original feedback transition.

The surviving Supervisor preserves the input ledger and original clocks. After old physical ownership ends, it advances the Owner namespace, replays the verified fact prefix, and cold-starts workers. Replay adds no inference; old publications gain no new authority. Both policies share this coordinator, durability, recovery, and actual Consumer.

c) Conditional continued service: For finite accepted work after finite supported failures, further progress requires

readable storage, eventual physical release/loading, terminating original calls that advance frontiers, fair Owner/Consumer scheduling, and sufficient service budgets. This is not an RTO or D2/C8 guarantee; loading, replay and recomputation retain their original clocks.

d) Control history: Roots, calls, buffers, visible views and outboxes pin their dependencies. Actual acknowledgment releases publication references, while the visible view stays pinned. Namespace/issue high watermarks prevent reuse after collection. Unreferenced terminal and audit windows have fixed limits; diagnostics stream to disk. These bound protocol entries relative to live dependencies, separately from current state, cumulative text and total memory.

V. Evaluation

A. Experimental design and service endpoints

We evaluate two service obligations under unchanged selected feedback histories: retaining dependency-valid progress already at the Consumer, and completing selected transitions before their original deadlines. The core factorial study isolates computation and whole-view retention; the native study tests input-confirmed full execution of the original correction pipeline. Timing, content support, and energy remain separate. The core experiments use actual neural calls, durable input and Canonical journals, and an instrumented FIFO Consumer. Measurements use Consumer applications rather than publication or display replay. Accepted input obligations retain their original arrival clocks.

a) Implementation and workloads: The service uses two 24-GB RTX 3090s: an FP16 Private worker and an FP32 Authority worker. Qwen3-ASR 1.7B [10] is the primary backend; Whisper base.en [11] tests a second adapter. The common carrier uses complete-row delivery, at most 16 rows per batch, a 256-token generation limit, the protocol kernel, and synchronized journals. These experiments use `row_complete`: physical early exit is disabled, so the effects of R and S cannot be attributed to cancellation. Model startup is outside the serving interval and is recorded separately. Serving energy integrates both physical boards; CPU energy is outside that measurement.

The independent confirmation comprises 24 previously unused speaker streams from LibriSpeech [12], totaling 714.11 s, selected using metadata before inference and partitioned into three eight-speaker cohorts. Its 110 cells contain 60 primary cells (three cohorts, two backends, two arrival patterns, five arms), 20 fresh-process repeat cells on cohort A, and 30 coarser-feedback controls. Primary Authority advances at every 2-s input boundary ($\rho = 1$); $\rho = 3$ changes both selection frequency and feedback segmentation. Synchronized streams start together; staggered streams start 0.25 s apart. Repeats and arms reuse these sources: the independent acoustic sample is 24 speakers, not 110 deployments.

b) Comparators: The four arms cross computation retention R with whole-view retention S (Section III). With R disabled, computation rebuilds from the selected root rather than retaining its witnessed continuation. With S disabled, a Canonical application replaces the provisional view with the new certified snapshot. This *checked-snapshot* Consumer still validates Private dependencies against the same root and passes the same protocol kernel. Removing retention does not remove validation, durability, input accounting, or the execution capability of Section IV. The fifth arm, *Full composition (ours)*, is a complete-composition positive control: an alternative policy instantiation with the same retention relation, Consumer protection, and shared execution/commit substrate. It is neither an external pre-existing implementation of our complete design nor an independently reimplemented neural stack. Comparable behavior is the expected positive-control outcome.

c) Endpoints: For offered interval i, let w_i be its audio duration, r_i its original ready time, d_i its first actual visible covering time, and c_i its first actual certified covering time. Original joint-deadline credit is

$$A_{2,8} = \sum_i w_i \, \mathbf{1}\{d_i - r_i \le 2 \ \wedge \ c_i - r_i \le 8\}. \qquad (5)$$

All offered audio, including end-of-stream (EOS) tails and misses, stays in its denominator. Whole-run service passes only when display P99 is at most 2 s and certified-view P99 at most 8 s. P99 is computed over offered frontiers within each run; credit weights their audio durations. Thus a P99 pass can coexist with a small tail miss. We separately measure adjacent visible-frontier withdrawals, their audio magnitude, and the integral of the deficit below the previously observed high watermark. Text revisions, final word error rate (WER), and board energy remain separate endpoints. Independent recounters check complete selected $F/Z/P/\mathrm{text}$ histories and reconstruct dependency witnesses from actual complete outputs and consumed inputs, rather than trusting a retention flag.

B. Does retained computation preserve delivered progress?

We test whether preserving valid work also preserves its presentation when selected updates replace snapshots. Table II reports the pre-specified Qwen, $\rho = 1$, staggered stratum, including all three cohorts. Retaining computation alone raises deadline credit from 654.115 to 664.150 s but increases audio-frontier withdrawals from 12 to 17. The additional work survives the new root, while snapshot consumption still discards its visible progress. Retaining views alone reduces withdrawals to one but yields 628.435 s of credit. The joint arm obtains 680.760 s and zero audio-frontier withdrawals: 26.645 s, or 3.73 percentage points of offered audio, above R_0S_0. Its gain over computation-only is 16.610 s.

The cohort-wise credit interaction, $I = Y_{11} - Y_{10} - Y_{01} + Y_{00}$, is $+18.530$, $+6.205$, and $+17.555$ s. These three contrasts share queues within each cohort. They support a composition effect in this stratum, rather than treating individual deadline windows as independent trials. The joint arm still erases 569 visible characters: zero audio-frontier withdrawal does not mean immutable text. Its 2.329-s median display P99 also remains above D2. The result preserves more timely progress without establishing that this workload meets its whole-run service target.

TABLE II
INDEPENDENT QWEN CONFIRMATION, $\rho = 1$, STAGGERED ARRIVALS. EACH ARM OFFERS 714.11 S ACROSS THREE EIGHT-SPEAKER COHORTS. WITHDRAWALS AND CREDIT ARE TOTALS; P99 AND ENERGY ARE MEDIANS OF THREE PER-RUN STATISTICS. ALL 15 RUNS MISS THE WHOLE-RUN D2 CRITERION.

Arm	Audio withdrawals	Display P99 (s)	D2/C8 credit (s)	Energy (J/audio-s)
R_0S_0	12	3.797	654.115	59.72
R_0S_1	1	3.891	628.435	59.16
R_1S_0	17	2.431	664.150	59.68
R_1S_1	0	2.329	680.760	59.37
Full (ours)	0	2.534	667.530	59.73

The full composition likewise has zero frontier withdrawals, the same 569 erased characters, and comparable timing and energy. Its lower credit than the joint arm in this stratum is not evidence that a different complete composition is inherently inferior. Across the confirmation, all 144 matched source/backend/granularity/arrival history groups agree on complete Canonical $F/Z/P/\mathrm{text}$; all 22 paired full-composition comparisons preserve those histories. The ablations therefore isolate how legal progress is retained, rather than buying latency by changing the selected feedback task. *Lesson:* computation and consumption protect different objects; their joint value is observed under the same selection contract.

C. Relative progress explains when retention matters

The useful opportunity is an already observed whole view ahead of a new Canonical frontier whose complete dependency path remains valid. In the headline joint arm, 18 of 368 Canonical updates encounter such a view and retain it. Computation-only encounters 17 and clears all 17. Under synchronized arrivals the joint arm encounters only three of 368; its credit is 0.790 s below computation-only. Without an ahead view, preserving it has little service value. These are witnessed incidences in the executed workload, not a production prevalence estimate.

Timing can also change the ranking. In the fresh-process staggered repeat on cohort A, joint credit is 226.320 s versus 230.565 s for computation-only, and $I = -2.245$ s. We retain this repeat alongside the three positive primary contrasts. The combination's value depends on relative progress and actual feedback compatibility; independent source confirmation does not imply an invariant timing advantage.

D. When does retention become sustained service?

We expect relative progress to determine both retention opportunities and deadline outcomes. We freeze three development workloads—eight synchronized sessions, eight staggered sessions, and sixteen staggered sessions—with four arms and three fresh processes per arm (36 cells). A single later exploratory midpoint adds twelve staggered sessions and the same twelve arm/repeat cells. Figure 3 retains all 48 runs. The nested source lists offer 121.205, 185.300, and 250.180 s per eight-, twelve-, and sixteen-session run. Adding sessions also adds source content; these are explicit operating workloads, not a pure concurrency capacity curve.

All arms pass whole-run D2/C8 in the two eight-session workloads. Joint display P99 is 1.426–1.528 s when synchronized, where the arms largely converge. Staggered joint P99 reaches 1.955 s, and an earlier development run reached 2.034 s, exposing a narrow timing margin. Two 0.215-s tails exceed D2 in the new eight-session runs and remain uncredited even though their runs' P99 passes. At sixteen sessions, all twelve runs fail D2, while C8 passes. Joint credit exceeds R_0S_0 by 20.280/22.885/20.180 s, but its credit interactions are $+20.910$, $+26.380$, and -9.575 s. Against computation-only, joint credit changes by $+18.910$, $+5.055$, and -7.225 s, and display P99 improves in only one repeat. Continuity is the stable distinction: computation-only withdraws compatible views 16/17/12 times, versus zero for joint.

First credit does not reveal a later withdrawal. A frozen secondary recount therefore follows intervals already delivered within D2 whose eventual certified covering view also meets C8. At each subsequent Canonical application it classifies withdrawal using the complete old view's dependency witness, separately recording necessary incompatible withdrawals. Loss lasts until an actual covering view returns or the next Canonical boundary censors this root's window. At sixteen sessions, computation-only still has 10/12/9 timely compatible loss windows with 24.903/23.635/15.736 audio-s×wall-s deficit area; joint has none. Such holes intersect the original D2 read point for 6.000/10.675/4.000 s of distinct audio. The secondary measures continuity after delivery; it neither changes Eq. (5) nor certifies provisional content.

The twelve-session witness is *not established*: all twelve runs fail D2, and computation-only never loses a timely complete non-EOS interval satisfying the frozen witness criterion. Joint has no compatible loss windows. The predeclared stop rule ends this search; full criteria and per-run results remain in the artifact. *Lesson:* compatible retention prevents a specific later loss; first-delivery performance and overload remain separate.

E. Quality and cost boundaries

All headline Qwen arms have WER $26/1908 = 1.36\%$ and identical selected histories. Retrospective support counting first-certified content plus first-ahead whole views that exactly match later same-F Canonical text leaves the headline credits unchanged. Elsewhere raw differences remain: joint loses 2 s of strict support in the sixteen-session third repeat and 2 s in each of the midpoint's first two repeats, although the frozen text normalizer agrees. This comparison is with the selected model output, not human-transcription correctness. Whisper exposes a larger adapter boundary: WER is 83.12% at $\rho = 1$ and 11.53% at $\rho = 3$, whose different feedback contract prevents interpreting it as a quality-preserving speedup.

The headline arms consume similar board energy (Table II); energy does not rank them in credit order. At sixteen development sessions, joint reduces total model time by 0.66–1.59 s against R_0S_0, while serving wall time does not consistently fall and the three energy differences change sign. The measured

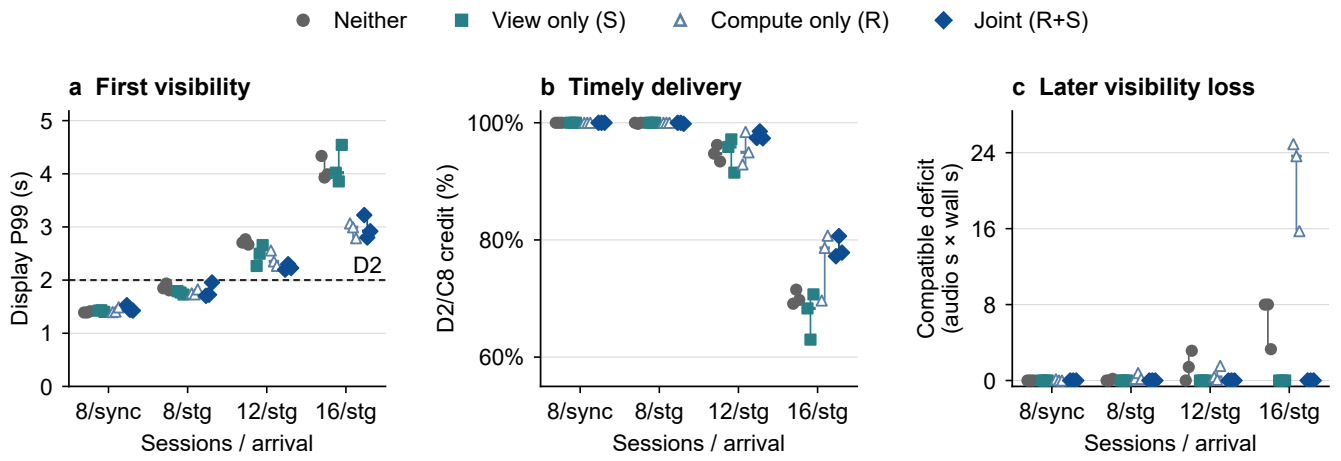


Fig. 3. First visibility, timely credit, and later visibility loss are different endpoints. All four workloads and three fresh runs per arm are shown, including the negative twelve-session midpoint. Points are real per-run values; ticks and ranges describe their median and observed range. Panel (c) is the separate secondary measure of subsequent compatible loss of originally timely intervals, censored at the next Canonical root when necessary. It does not replace panel (b)'s original first-observation credit.

benefit is the progress delivered and preserved under a fixed feedback contract. It is not a general reduction in inference cost or a cure for overload.

F. Do early full calls preserve the original task?

We test whether input-confirmed calls improve service while retaining original learned correction and recursive feedback. The StreamCorrect adaptation uses fixed 500-ms inputs, no VAD, and original FP32 execution [1]. Both author-built policies have two complete workers, the same ASR early-view Consumer and neural warmup. Conservative execution waits for selected parent feedback; Full-call permits conditional execution. We distinguish the earlier implementation (`bdfa`) from the final resilience implementation (`4f`); full execution identities are in the artifact.

a) Independent confirmation and matched regression: The earlier method was frozen before selecting four unused AISHELL-1 speakers [13] by metadata in an 8–9-s band. Two policies and two fresh repeats give 16 runs with randomized pair blocks, reversed arm order and swapped GPU roles. All selections, 500-ms frontiers and partial EOS remain. Speakers, rather than repeated runs, are the acoustic units. All 272 selected transitions match complete original start/end states, correction and returned IDs against offline full-call references. Both arms have five edits over 81 unique reference characters (6.17% CER); this is the adaptation's accuracy, not the published benchmark. Earlier Full-call raises credit from 37.0 to 66.33975 s and passes from 0/8 to 8/8.

The matched 32-run study holds those four sources, FP32 models, warmup, allocator and external board-power observation fixed across versions. All 544 selected transitions match complete original inputs/outputs, correction and IDs under the declared static code-version mapping. Final Full-call earns 65.821625 versus 30.0 s and passes 6/8 versus 0/8 (Table III).

TABLE III
FINAL FULL CALLS PRESERVE TASK FIDELITY AND IMPROVE TIMELY SERVICE. THE MATCHED VERSION STUDY REUSES FOUR MEASURED SPEAKERS: TWO POLICIES, TWO VERSIONS AND TWO FRESH TRIALS (32 RUNS). EACH ROW OFFERS 66.340 S; CREDIT AND ENERGY ARE TOTALS, D99 IS THE MEDIAN OF PER-RUN P99. ENERGY USES A COMMON EXTERNAL ORIGIN-TO-CONFIRMED-PHYSICAL-MODEL-DRAIN WINDOW. THIS IS REGRESSION EVIDENCE, NOT NEW ACOUSTIC CONFIRMATION.

Version	Policy	Credit (s)	D99 (s)	Energy (kJ)	Pass
Earlier	Conserv.	33.500	4.450	44.690	0/8
Earlier	Full-call	66.340	1.633	39.735	8/8
Final	Conserv.	30.000	4.759	47.019	0/8
Final	Full-call	65.822	1.638	43.952	6/8

Its two failures are S0908 EOS: display P99 increases by 0.163/0.050 s against Earlier Full-call. Other paired differences have both signs; similar medians do not establish equivalence. Final Full-call uses 6.52% less drain-window board energy than Final Conservative, but 10.61% more than Earlier Full-call. Model drain and source-specific cleanup are different endpoints; EOS-window and original source endpoints remain separately available.

b) Resident independent requests: Models load once per trial and both worker PIDs persist across four pre-registered independent requests. The next request begins only after actual certified consumption of the previous file EOS. This is single-outstanding closed-loop service, without online VAD or open-arrival capacity. Two policies and two fresh trials retain all 272 original transitions and each request's own input clock. Full-call passes D2/C8 on 7/8 requests versus 0/8, earning 64.118875 versus 31.0 of 66.33975 s. The second S0908 request has D99 2.229 s; residency does not remove its deadline boundary.

c) What content arrives, and what remains?: The earlier independent bank contains 92 promoted calls beginning before their selected parent finishes; 20 have wrong own predicted ends. Twenty-four invalid/covered calls stop after ASR and two complete wrong-input results are discarded. Input confirmation, rather than prediction correctness, explains which original work survives.

A frozen zero-neural recount of the final matched and resident logs checks actual complete dependencies and first-view text at its own frontier G. Credit whose whole first view retrospectively equals selected text at that same G rises from 22.5 to 51.0 s in matched runs, and from 24.0 to 50.259 s in resident requests. Full-call's remaining 14.822/13.860 s mismatches; unknown coverage is zero. Raw and whitespace comparisons agree. Each arm in either bank introduces 180 code-point occurrences, retracts 14, and leaves 166 last-visible. All-offered whole-text matches are 106/136 for Conservative and 104/136 for Full-call. The gain is timely supported content, not more text or a higher overall match fraction; agreement with selected model output is not human accuracy.

The final matched bank has no compatible-ahead application, so gives no retention-rate evidence. Resident Full-call has one nonempty opportunity: its 5.0-s whole view survives a compatible 4.5-s selected update with the same tip and job. The complete tip is logically held for 0.817 s spanning that update before legitimate supersession; this is Consumer event-state observation, not avoided downtime. Each Full-call bank contains two necessary incompatible withdrawals. Final objects are right-censored at the last actual same-source observation. *Lesson:* confirmed full calls accelerate the original task, while content support, continuity opportunities and deadline margin require their own denominators.

G. What survives a producer or Owner failure?

We test whether failures preserve acquired results without granting old messages new authority. The final implementation shares models, durability, guards, recovery and Consumer across both policies. Initial loading and fixed warmup precede service; replacement loading, replay, recomputation and physical drain retain the original clock.

Six conditions, two policies and two fresh processes on one development clip give 24 short trials, plus four EOS/compatible-view regressions. With eight sustained-state trials below, the bank has 3,976 durable transitions independently bound to original complete receipts and actual Consumer applications: zero illegal installations or incorrect owned-message rejections. Eight duplicate/late messages cause no selected mutation; eight worker deaths have confirmed physical replacements. Conditional calls supply 354 durable facts: 352 have original Owner installation records, and two first reach the Consumer through recovery. In four worker-after-ASR cases, an already consumed whole view remains across producer exit/fencing until legitimate correction. This is retained Consumer state, not redelivery or proof that the prediction was correct.

Table IV keeps all deadline failures. Normal and duplicate Full-call each pass one of two trials; the other misses D2 only at EOS. All worker-fault trials fail D2 but pass C8; Owner-fault trials fail both. Old durable facts reappear in 1.36–1.81 s, while new certified post-fault completions take 64.61–108.08 s in Owner cases. A surviving worker may finish before replacement readiness. Old-fact replay and new inference are separate recovery endpoints.

a) Complete buffers across producer death: The bank above never installs a completed conditional buffer. Four frozen timing controls cover two matching and two mismatching inputs. A transport fixture holds the parent after original ASR and the child after sending its complete original result. After the Owner registers the result, we kill its producer, verify exit, CUDA release, sealed-channel validation and fencing, then release original parent correction. Both matching results install and are consumed once with their original job, birth and incarnation; both mismatches are rejected against the actual same-frontier parent. Positive cases also finish a subsequent independent request. These are controlled interleavings, not natural incidence or fault-time SLO wins. A separately frozen zero-neural reconstruction validates the unchanged runs after correcting a READY indexing defect; original failed audit gates remain in the artifact. *Lesson:* producer failure ends future message authority without uniformly invalidating accepted results.

b) Sustained-state stress and its limits: Two metadata-selected AISHELL-1 speakers contribute 24 utterances each, concatenated as exact PCM into 228.51025/208.1003125-s streams without join resets. A continuous original-engine reference and 875 separately restored steps agree on complete state, correction and IDs, crossing 398/357 buffer shifts and 353/252 context changes. Eight serving trials produce 3,500 matching selected steps. Unpinned terminal and each audit class stay at most 64 entries; live dependencies, state/output bytes, Owner RSS and cumulative logs are separate.

All eight trials fail D2/C8 (display P99 449.727–502.952 s). Both policies reproduce severe original-task insertion errors: CER 3103/486 (638.48%) and 1313/522 (251.53%). Full-call uses more energy. The original length guard can skip decoding, so frontier credit need not add text. These are state/collection stress tests, not deployable continuous ASR. Across all 36 trials, the checker preserves 90 compatible-ahead dependency-tip applications and records 1,118 necessary incompatible withdrawals. Internal tips need not add text; these controlled counts do not estimate natural prevalence.

An independent dispatch audit finds 360 eventual matches among 1,901 conditional inputs; 1,541 are refuted, none unresolved. Six matched jobs never become durable after injected failures. Workload-specific ratios remain separate. Original-clock misses and a conservative offline excess-length bound precede the after-trim 448-token condition on each long stream; neither yields online prefix CER or a unique causal diagnosis.

H. From physical release to supported service

A separate 24-run core study crosses R and K with S enabled, two arrival patterns and three fresh processes, using

TABLE IV

FAULT CONTAINMENT PRESERVES LEGALITY, WHILE COLD RECOVERY MISSES THE ORIGINAL SERVICE DEADLINES. SIX SHORT CONDITIONS USE ONE FIXED CLIP AND TWO FRESH TRIALS PER POLICY. EACH CELL LISTS CONSERVATIVE ABOVE FULL-CALL; RANGES ARE OBSERVED MINIMA/MAXIMA. ILLEGAL; DONE GIVES ILLEGAL INSTALLATIONS AND COMPLETED/2 TRIALS. REJECT/MUTATION COUNTS LOCAL REJECTIONS AND SELECTED CHANGES. NEW RESULT IS FAULT-TO-ACTUAL-CERTIFIED CONSUMPTION OF A NEW FACT WHOSE CALL COMPLETES AFTER THE FAULT, EXCLUDING OLD-FACT REPLAY AND REPLACEMENT READINESS. D2/C8 COUNTS JOINT WHOLE-RUN PASSES; EOS COUNTS D2 MISSES. BOTH-BOARD ENERGY RUNS THROUGH PHYSICAL DRAIN.

Condition	Illegal; done	Reject / mutation	New result (s)	D2/C8 pass	EOS miss	Energy (J)
	0; 2/2	0 / 0		0/2	2	6566–6574
No fault	0; 2/2	0 / 0	–	1/2	1	5896–5911
	0; 2/2	4 / 0		0/2	2	5701–5893
Duplicate / late	0; 2/2	4 / 0	–	1/2	1	5538–5846
	0; 2/2	0 / 0	1.90–2.22	0/2	2	5182–5401
Worker after ASR	0; 2/2	0 / 0	2.15–2.21	0/2	2	5447–5647
	0; 2/2	0 / 0	2.10–2.25	0/2	2	4775–5178
Worker partial frame	0; 2/2	0 / 0	1.76–2.23	0/2	2	5492–5594
	0; 2/2	0 / 0	86.42–108.08	0/2	2	24549–28019
Owner before durable	0; 2/2	0 / 0	64.61–107.09	0/2	2	19571–26557
	0; 2/2	0 / 0	83.90–87.05	0/2	2	22415–24200
Owner after durable	0; 2/2	0 / 0	101.80–104.44	0/2	2	26483–27171

a frozen wrong-conditioning fixture. Selected histories and WER $14/687$ match; all 675 same-input/prompt/audio/peer-order pairs return identical outputs. Synchronized $R1$ releases the next 16-row computation 473–577 ms earlier. All 96 following rows are actually consumed and their whole texts match later same-frontier Canonical text. Only 6/6/1 rows switch from deadline miss to hit. Adding K changes credit by +7.970 to +15.255 s when synchronized and -26.000 to $+21.780$ s when staggered, without consistent energy savings. Request acknowledgment still precedes physical drain. This is a measured release-to-service opportunity, with no stable positive $R \times K$ interaction.

VI. RELATED WORK

a) Incremental information and dependency reuse: The incremental-unit model separates newest committed information from the active right edge and propagates dependency revocation [2]. Its commit promises that a unit will not be revoked. Here, certification records acceptance by the protected selection protocol; later selected facts may still revise the transcript. Adapton checks recorded dependencies; Nominal Adapton combines names with arguments and dependency checks to recover reusable work [3], [4]. Differential Dataflow maintains incremental computations across evolving versions [14]. These works establish much of the basis for dependency-aware reuse. RevTrace instantiates that basis for complete neural feedback state and frozen executed calls. Its service obligation connects execution eligibility to the actual whole view retained when an independently selected root is applied. Our sufficient witness follows recorded executions; it does not transfer a from-scratch theorem to independently repeated floating-point inference.

b) Revisable speech services and speculation: Whisper-Streaming converts an offline recognizer into incremental service using LocalAgreement [15]. StreamCorrect adds learned correction and feeds selected corrections into subsequent recognition [1]. RevTrace retains that selection and feedback behavior in its controlled native adapter; its contribution concerns the asynchronous execution and consumption around the feedback boundary. It does not introduce a better acoustic model or train a new corrector.

SSBD coordinates old-output draft reuse with display-only suffix masking [5]. It revalidates a draft under updated input and includes an unbiased configuration. RevTrace follows complete calls that actually executed to establish their eligibility under newly selected feedback state, then checks the dependency witness for the Consumer's whole current view. PredGen provides a close precedent for speculation while speech input evolves across recognition, language generation, and speech synthesis [6]. Our comparison holds the complete selected feedback task fixed across the retention policies. The specific reuse object, protected selection, and whole-view transition define the composition evaluated here. RelayS2S uses a learned verifier to commit a fast-path prefix that constrains subsequent slow-path generation [16]. Our native calls instead confirm their complete inputs and retain the original correction and feedback update. Work on ASR alignment drift also establishes the importance and cost of audio position state [17]; our contract includes that state rather than treating text equality as sufficient.

c) Execution, authorization, and recovery: Retire jointly addresses revocation, valid state inheritance and physical reclamation for runtime-selected version transitions [18]. RevTrace derives complete-call eligibility and whole-view actions from an unchanged selected feedback task: an unsupported view can be withdrawn while a future call's input remains unresolved. Our covered-exit policy is execution support, without a new general cancellation claim. The parent-version control does not reproduce the complete Retire system.

Optimistic and multiversion concurrency control separate validation and commit from concurrent execution [19], [20]. Write-ahead recovery persists state changes and identities [8], and stream-processing systems coordinate recoverable state [21]. We use these established principles at the ASR feedback boundary. They supply machinery for ordering and

durability; the adapter still has to specify which state constitutes a valid continuation and which complete view the Consumer may retain.

VII. Discussion

a) Scope of the guarantee: The continuation witness assumes a sufficient adapter state and faithful execution of frozen inputs. A new adapter must account for every mutable field that can affect later inference, selection, or feedback. A hidden KV cache, alignment cursor, or external tool state can invalidate a token-only argument. Conservative rejection remains allowed when a sufficient witness is absent. The guarantee concerns dependency validity and an individual compatible application; it supplies neither transcription correctness nor monotone text for an entire session.

b) Progress, content, and capacity: Processing more audio can leave the cumulative transcript unchanged. Frontier credit, actual text additions and revisions, eventual selected text agreement, and ground-truth accuracy answer different questions. We report them with their original denominators. Preserving compatible progress also cannot remove all input synchronization, queueing, or neural work. The negative c12 witness identifies a timing boundary. An earlier native two-source test earns 17.0 s with retention versus 18.5 s for independent full replicas, at higher energy and a different warmup/Consumer. These do not define a universal capacity envelope. Independent native confirmation establishes a single-chain service opportunity with the original selected task; it does not turn that result into saturated multi-source capacity.

c) Deployment and validation: The continuous-original reference exercises the native cut across real buffer and context changes; another adapter must establish its own state sufficiency. Active-dependency collection bounds protocol entries relative to live references, separately from cumulative text, current state, and total memory. Owner/worker recovery assumes the surviving services of Section II. Replaying old facts restores access to that prefix; the measured cold Owner restarts still miss original deadlines for new work. Broader ownership transfer requires separate validation. Finite specifications, source controls, controlled faults, actual GPU service, and retrospective checks provide distinct evidence.

d) Eligibility and resource admission: A valid input witness determines whether an owned call may be used. It does not determine whether starting another conditional call is worthwhile. Limiting new speculation can leave existing supported whole views intact. The long stress study motivates this separate resource decision. In a bounded two-client probe, new speculative calls precede another client's ready necessary call in both repeats. The later client has no additional C8 misses, and its C99 penalty fails the predeclared repeat-consistency gate. No admission controller or general native-session fairness is evaluated.

VIII. Conclusion

RevTrace separates complete-input eligibility, original selection authority, and whole-view support while trusted feedback advances. Service experiments expose a coordination gap: first delivery can meet its deadline, yet a compatible update can withdraw that progress. The joint design protects the supported ahead view. Native full calls can retain original correction work after their inputs are confirmed even when their own next predictions are wrong. Durable facts and physical identities carry this distinction through supported process failures. The resulting composition connects dependency reuse to actual continued consumption under a fixed feedback task. Computation-only retention increases frontier withdrawals in the headline factorial stratum; joint retention prevents them. Final native calls improve timely supported-content credit while keeping necessary revisions and EOS misses. Actual support, timing, recovery and capacity bound the composition's benefit. Extending recovery to Supervisor failure remains a separate direction.

AI Tool Usage

ChatGPT/Codex assisted literature discovery, drafting and revision of all manuscript sections and the refinement argument, runtime extensions and experiment orchestration, and validation/analysis code producing the figures and tables from recorded measurements. The artifact records execution sources, frozen configurations, controls, and result provenance, separating actual service measurements from component tests and retrospective analyses. Constructed streams concatenate recorded AISHELL-1 PCM. The authors retain responsibility for the submission.

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